ENTROPY-BASED ANALYSIS OF ENCRYPTED DATA

Mini Project-1 · P6 · Group G6

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INTRODUCTION

Entropy is a measure of uncertainty or randomness in data. In cybersecurity, it helps us study how predictable data is and understand the characteristics of encrypted information.

WHAT THIS PROJECT EXPLORES

We calculate and compare entropy values to study the statistical characteristics of encrypted data and find out how effective entropy is in deciding the security of an Encryption using shannon entropy

PROBLEM

The Technical Problem

Scenario: an analyst must study encrypted and transformed data samples without decrypting them.

Inputs: text, PDFs or any file.
Outputs: byte frequencies, entropy, comparisons and a verdict.

Question: can these statistics tell kinds of data apart, and what can they not conclude?

WHY IT MATTERS

Why Entropy Matters for Encrypted Data

Higher entropy indicates less predictable ciphertext, making patterns harder to detect.

Randomized encryption helps hide relationships between repeated messages.

Entropy analysis helps identify predictable patterns and compare encryption outputs.

Shannon Entropy and Its Formula

$$H(X) = -\Sigma\, p(x) \log_2 p(x)$$

H(X)

Entropy

Overall unpredictability of the data, in bits

p(x)

Probability

How often symbol x shows up in the sample

log 2

Information

Bits of“surprise”carriedby one occurrence

Σ

Sum

Technical Implementation:Entropy Analysis of Encryption

Method

- Transform identical input with 8 hand-written methods (no cryptography libraries)
- Compute Shannon entropy before and after; compare via ΔH

Model

- $p(x)$ = count of byte value x ÷ total bytes (256 possible values)
- $H(X) = -\Sigma\, p(x) \log_2 p(x)$
- $\Delta H = H(Y) - H(X)$

Example: Caesar Cipher (shift = 3)

- HELLO → KHOOR; $H(X) = H(Y) \approx 1.922$ bits/char
- Substitution preserves frequencies, so entropy is unchanged

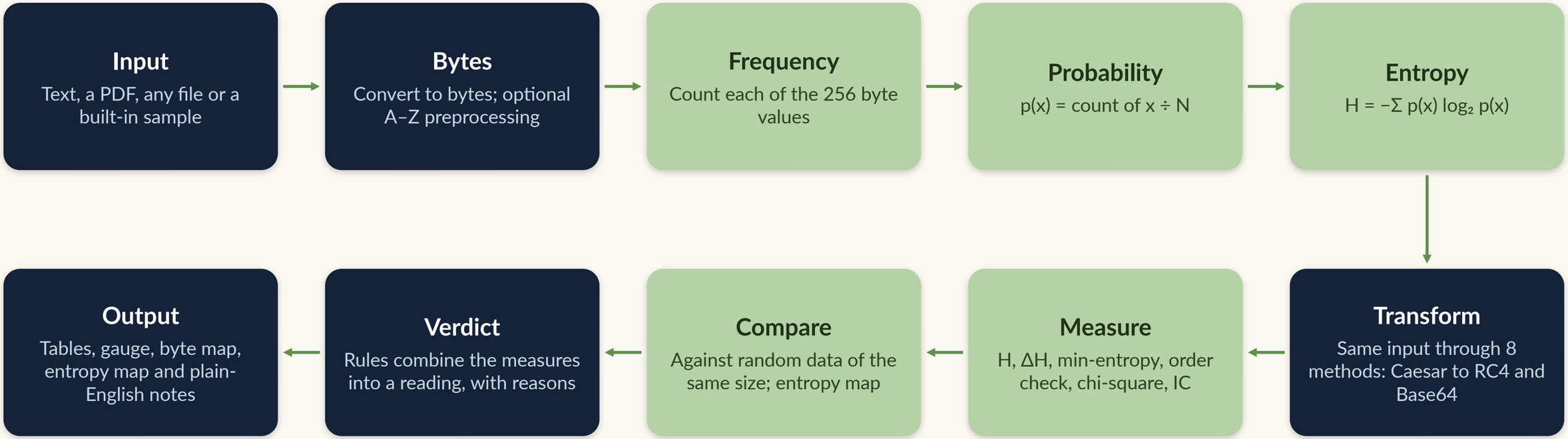
Takeaway

- Entropy measures byte-level unpredictability, not cryptographic strength

Proposed solution

A menu-driven Python application for entropy-based analysis of data, including encrypted content. Users load text, a PDF or any file, count byte frequencies, measure Shannon entropy and compare eight hand-written transformations: Caesar, Vigenère, Rail fence, XOR, one-time pad, RC4, Base64 and a random reference. Extra measures, an entropy map and an automatic verdict show how far statistics can tell structured, encoded and random-looking data apart, and where entropy fails as a security metric.

Solution Flow



The experiments (sample size, mixing in noise, building a distribution) reuse the same frequency, probability and entropy steps.

Core technical components

- Python 3 standard library only: math, sys and os
- Menu-driven console application (main.py)
- All maths in entropy_core.py, written by hand
- No cryptography libraries, no ready-made counters
- Caesar, Vigenère, Rail fence, XOR, one-time pad, RC4
- Hand-written Base64 encoder and random generator
- Min-entropy, order check, chi-square, coincidence index
- Optional: Flask web dashboard, pypdf for PDF text
- Unit testing with unittest: 37 known-answer tests

HOW IT WORKS

A Menu-Driven Application

Look at the data: byte frequencies, a 16 × 16 byte map, an entropy map and a verdict.

Experiment: the same input through 8 methods, sample size, and changing the distribution.

Understand: the limitations, a 9-step guided demo and a glossary of every term.

EXTRA MEASURES

Beyond Shannon Entropy

Min-entropy: $-\log_2$ of the most common byte's probability. The worst-case guess.

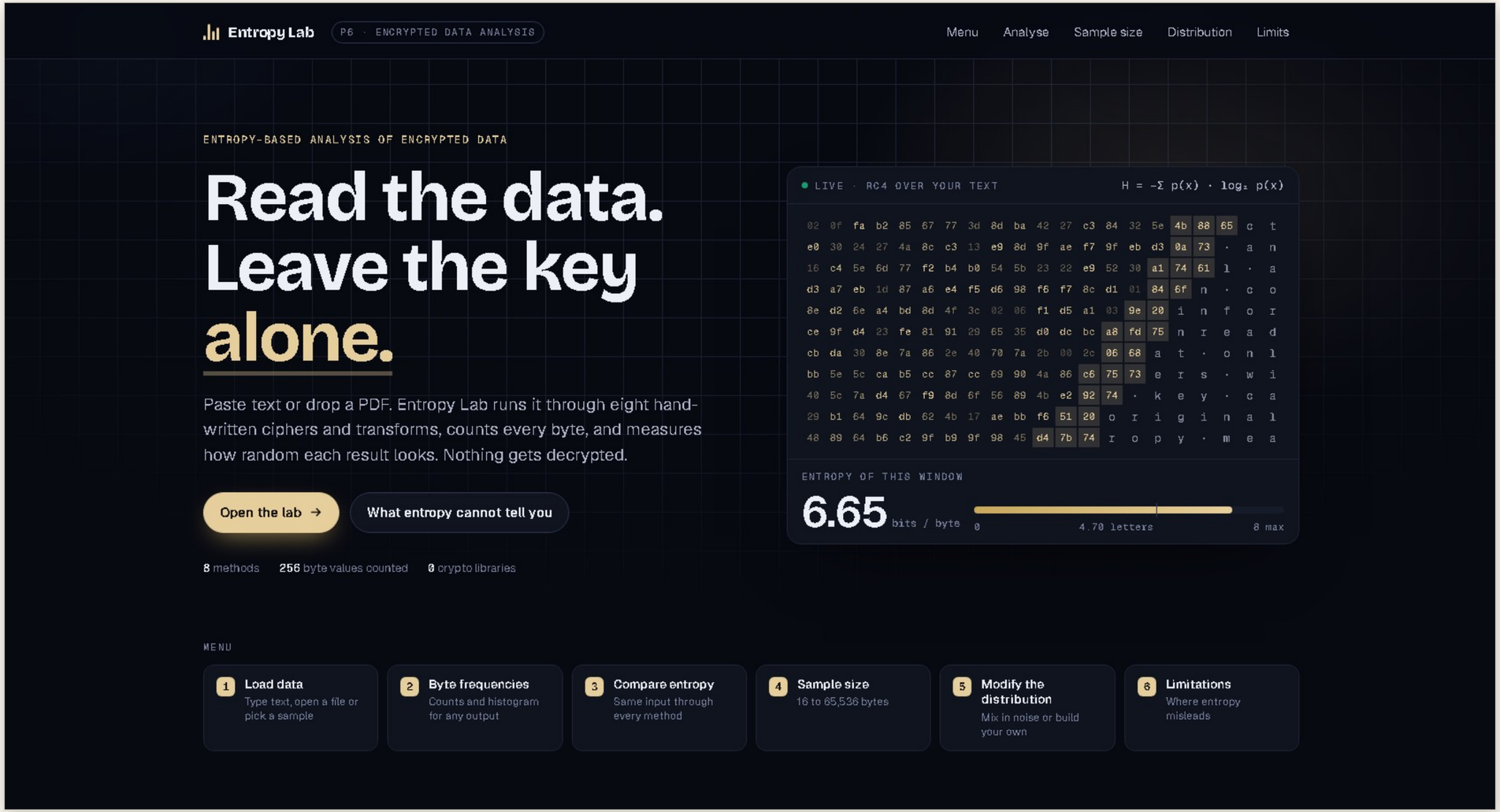
Order check: entropy of the steps between neighbouring bytes. Catches sorted and counter data.

Chi-square: are the counts as even as chance allows? Random data scores between −3 and +3.

Index of coincidence: English 0.067, random letters 0.038. Tells Caesar from Vigenère.

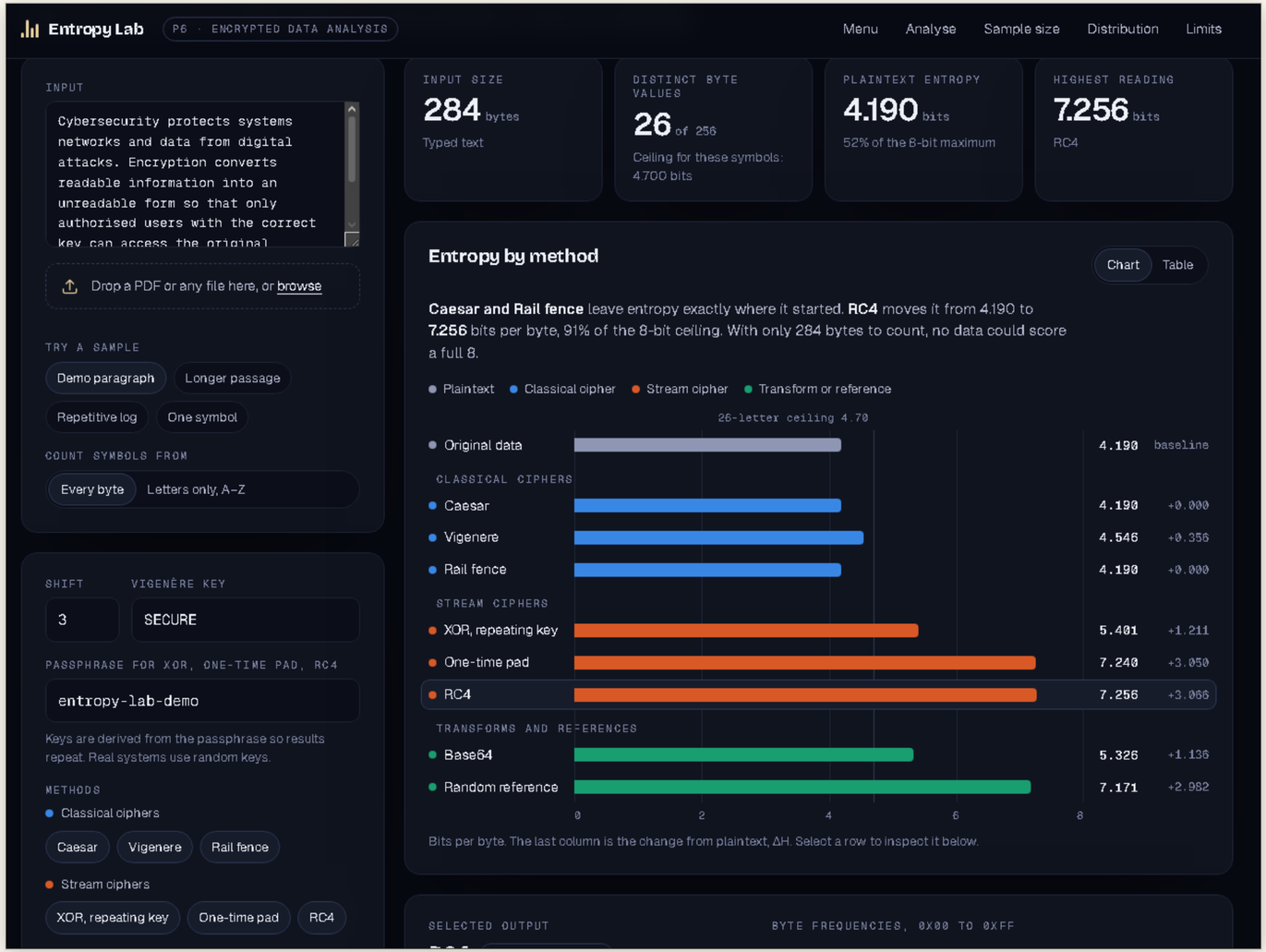
Demo: Home Screen and Menu

RC4 encrypts the text live while the window entropy updates. The numbered menu leads to every part.



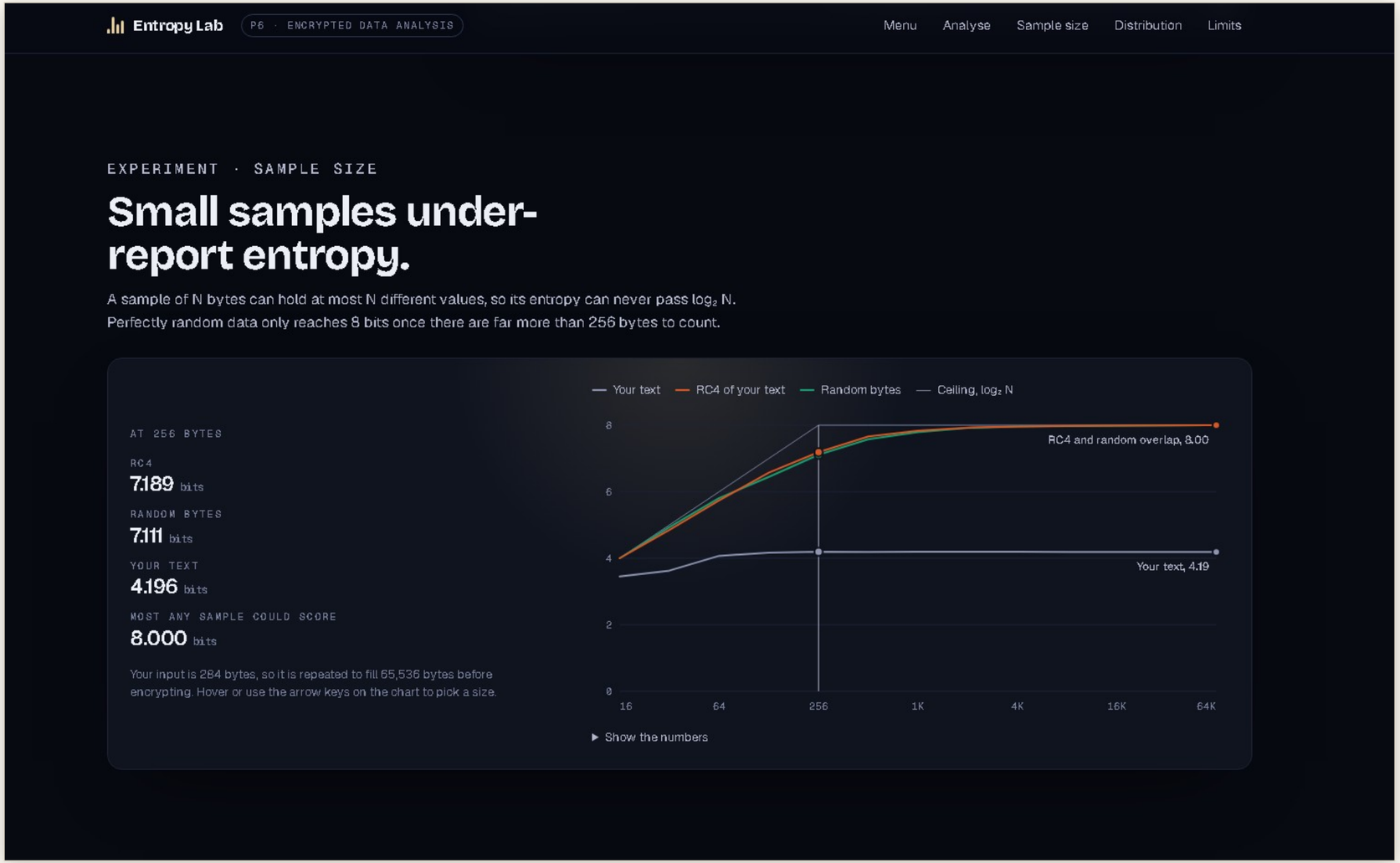
Demo: Entropy by Method

One input, eight methods. Caesar and Rail fence stay at 4.190; RC4 lifts entropy to 7.256.



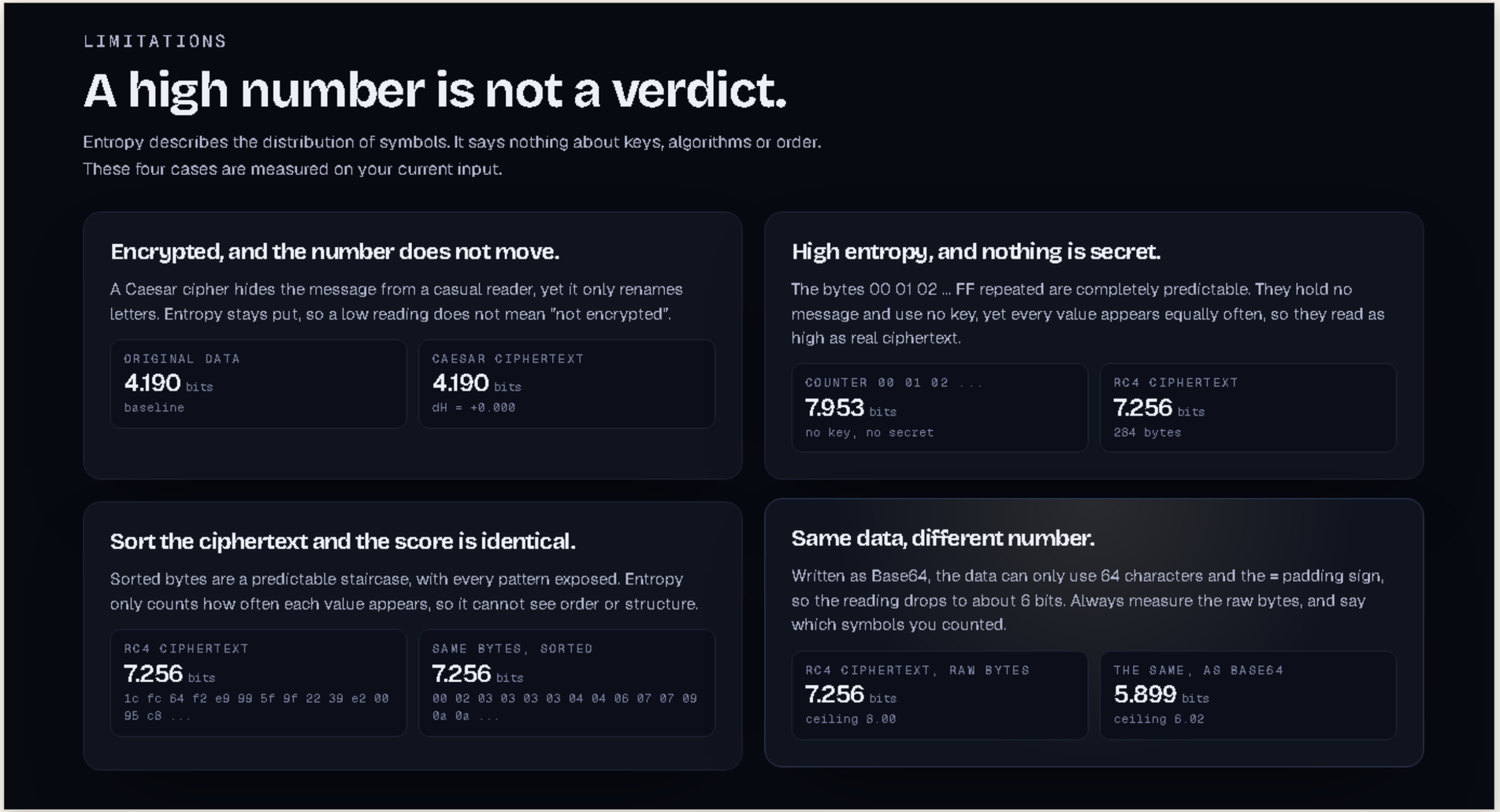
Demo: Sample-Size Experiment

At 256 bytes RC4 reads only 7.19. By 65,536 bytes RC4 and random data both reach 8.00.



Demo: Limitations, Measured

Four cases on the same input where an unchanged or high entropy reading misleads.



ADVANTAGES

Advantages of Our Approach

Statistics only: any file is analysed byte by byte without decrypting it, as the brief requires.

Fair comparisons: every reading is checked against random data of the same size.

More than one number: the order check and chi-square catch sorted and counter data.

Entropy map: shows where encrypted data sits inside a file, not just one average.

LIMITATIONS

Limitations of Entropy Analysis

Fixed thresholds: our verdict bands and the 6.3-bit map cut-off were chosen by us, not derived.

Cannot tell encrypted data from compressed or random data, even with all the measures.

Small samples: under about 1,000 bytes readings are unreliable; chi-square needs 1,280.

Teaching ciphers with seeded keys: results repeat, but say nothing about real key strength.

INDIVIDUAL CONTRIBUTIONS(Primary roles)

Anupama : Mathematical Modelling

Hari:Implementation and coding

Karthika:Algorithm design

Nisha:Presentation and technical communication

Rohit:Project coordinator and Data experimentation

Sreerag:Documentation

Vandana:Testing and Validation

Conclusion

- Entropy separates structured data from random-looking data, but cannot identify encryption
- Caesar and Rail fence hide text without changing entropy; RC4 pushes it close to 8
- Small samples read low, so compare with random data of the same size
- Extra measures and the entropy map tell more data types apart; none prove security

GitHub repository

github.com/harishankarsa454-code/G6-Entropy-Analysis

Questions?

